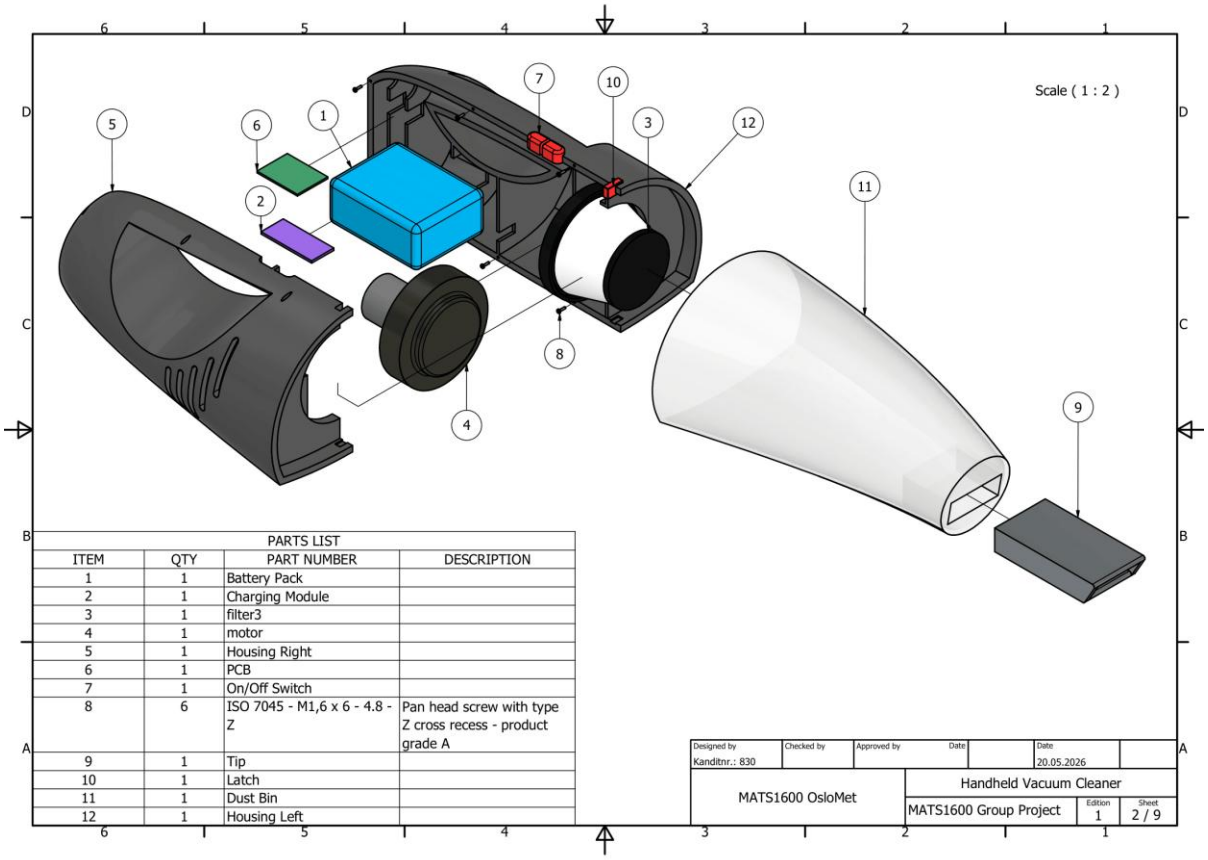
Handheld Vacuum Project Report

Student numbers	886, 830, 878, 866, 899
Group	1
Report submission date	22-May-2026



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	Battery Pack	
2	1	Charging Module	
3	1	filter3	
4	1	motor	
5	1	Housing Right	
6	1	PCB	
7	1	On/Off Switch	
8	6	ISO 7045 - M1,6 x 6 - 4.8 - Z	Pan head screw with type Z cross recess - product grade A
9	1	Tip	
10	1	Latch	
11	1	Dust Bin	
12	1	Housing Left	

Designed by	Checked by	Approved by	Date		Date	
Kanditnr.: 830					20.05.2026	

MATS1600 OsloMet	Handheld Vacuum Cleaner		
	MATS1600 Group Project	Edition 1	Sheet 2 / 9

Contents

Handheld Vacuum Project Report 1
1. Introduction 4
1.1 Brief 4
1.2 Product Specifications 4
1.3 Project Planning 5
2. Research 5
2.1 Market and competitor analysis 5
2.2 Customer analysis 6
2.3 Background Knowledge / Theory 6
2.4 Cross-Domain Analogies 7
2.5 Standards and Technical References 7
3. Concept Development and Selection 8
3.1 Ideation Technique 8
3.2 Concept Selection Process 9
3.2.1 Overall concept selection 9
3.2.2 Charging interface selection 10
3.2.3 Battery configuration selection 10
4. Detailed Design 11
4.1 Design Schematic 11
4.2 Choice of Standard Components 11
4.2.1 Motor Selection 11
4.2.1.1 Choice of Brushed Motor: 12
4.2.1.2 Suction estimation: 13
4.2.2 Battery Selection 14
4.2.2.1 Battery Protection (PCM) 15
4.2.3 Temperature Monitoring Component Selection 15
4.2.4 Charging Module Selection 16
4.2.5 Electrical Circuit Selection 17
4.2.6 Filter Selection 17
4.3 Technical Design of Unique Components 18
5. Validation of Design 22
5.1 FMEA 22

5.2 Testing / Validation 23
6. Evaluation of Final Design 23
7. References 24
8. Appendix 27

1. Introduction

1.1 Brief

The objective of this project is to design a compact and portable handheld vacuum cleaner for quick cleaning tasks in homes, offices, and vehicles. The product should be cordless, easy to handle, and suitable for light to moderate cleaning such as dust, crumbs, and small debris. The design should balance suction performance, battery runtime, filtration, safety, charging accessibility, and cost-effectiveness.

1.2 Product Specifications

The following specifications define the target requirements for the handheld vacuum cleaner. The specifications are listed in order of priority.

1. **Usability and Convenience**
    - Cordless operation
    - Lightweight and compact design
    - Easy handling and intuitive controls
2. **Battery Performance**
    - Minimum runtime: **≥ 20 minutes under normal operation**
    - Extended runtime mode: **≥ 40 minutes**
    - Maximum charging time: **≤ 3 hours**
3. **Cleaning Performance**
    - Minimum target suction pressure: **≥ 3000 Pa**
    - Effective removal of dust, crumbs, and small debris from hard surfaces, furniture, workspaces, and car interiors
    - Airflow path should minimize pressure losses through good sealing and limited sharp transitions
4. **Filtration Performance**
    - High-efficiency filtration of fine dust particles
    - Replaceable filter element
5. **Safety**
    - Safe operation during both use and charging
    - Protection against overheating, overcharging, and short circuits
6. **Charging and Accessibility**
    - Charging solution must be simple and widely accessible
    - Compatible with commonly available power sources
7. **Form Factor**
    - Compact internal layout suitable for handheld use
    - Components must fit within a portable enclosure
8. **Cost**
    - Designed for cost-effective production and a broad consumer market

1.3 Project Planning

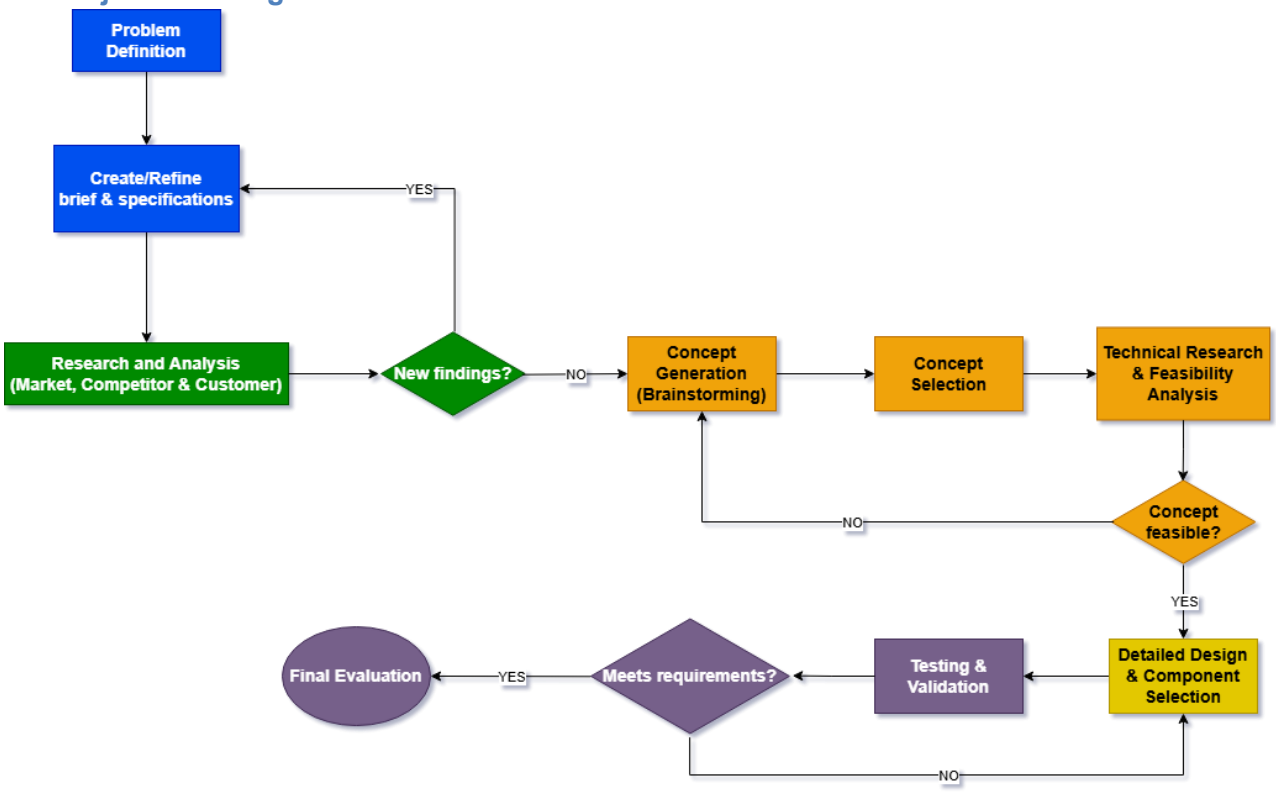


Figure 1: *Engineering design process used in the project.*

Figure 1 shows the engineering design process used in the project. The process started with defining the brief and specifications, followed by research, concept generation, concept selection, detailed design, and validation. If a concept or component choice did not meet the requirements, the design was refined before the final evaluation.

2. Research

2.1 Market and competitor analysis

The global vacuum cleaner market is mature, highly competitive, and continues to grow steadily. The market is projected to increase from USD 11,150 million in 2020 to USD 13,840 million by 2026, corresponding to a compound annual growth rate (CAGR) of 3.7% (Valuates Reports, 2021). This growth is driven by increasing purchasing power, rising health and hygiene awareness, and a growing working population.

A key trend in the market is the increasing demand for compact and convenient cleaning appliances. Cordless and handheld vacuum cleaners are particularly valued for their portability, ease of use, and ability to perform quick cleaning tasks across multiple environments (Fortune Business Insights, 2024; Valuates Reports, 2021).

Within this segment, products are typically differentiated by size, weight, battery performance, and overall usability. Existing solutions generally reflect a trade-off

between performance and cost. Premium products emphasize high suction power and advanced technology, while lower-cost alternatives prioritize simplicity and compact design, often with reduced runtime or performance.

These trends indicate a market opportunity for products that balance sufficient cleaning performance with high usability and accessibility. This is particularly relevant for general-purpose handheld vacuum cleaners intended for everyday use.

2.2 Customer analysis

The primary customers in the vacuum cleaner market are private households, where ease of use and convenience are key decision factors. Handheld vacuum cleaners are typically used as a supplement to a main vacuum and are intended for quick, localized cleaning tasks.

Customers in this segment generally prioritize compact size, low weight, and simple operation over maximum suction performance. Products that enable fast and effortless cleaning are therefore perceived as highly valuable.

Consumer research further indicates a clear willingness to pay for convenience. A global study by GfK shows that 41% of consumers are willing to pay more for products that simplify everyday tasks and save time (GfK, 2024, p. 16). This highlights the importance of features that reduce effort and increase usability.

In addition, innovation that targets specific user needs is identified as an important factor for attracting consumers, particularly when it improves convenience and aligns with changing lifestyles (GfK, 2024, p. 13).

Based on these insights, the target user can be defined as a general consumer seeking a practical, accessible, and easy-to-use cleaning solution for everyday environments such as homes, offices, and vehicles.

2.3 Background Knowledge / Theory

A handheld vacuum cleaner works by using a motor-driven fan to create a pressure difference between the nozzle and the exhaust. This pressure difference causes air and dust particles to flow through the nozzle, dust chamber, filter, fan, and exhaust path. The useful air power can be described by the relationship:

$$P_{air} = \Delta P \cdot Q$$

where P_{air} is air power, ΔP is pressure difference, and Q is volumetric airflow rate. This relationship shows that cleaning performance depends on both suction pressure and airflow rate, rather than motor power alone (Engineering ToolBox, n.d.-a).

In practice, part of the available power is lost due to friction, turbulence, leakage, filter resistance, bends, and sudden changes in cross-sectional area. These losses are

important in a compact vacuum cleaner, where small ducts and sharp transitions can significantly reduce airflow. Therefore, the internal airflow path should be smooth, compact, and well sealed, especially around the dust chamber and HEPA filter (Engineering ToolBox, n.d.-b).

Battery runtime depends on the energy stored in the battery and the electrical power consumed by the motor. Battery energy can be estimated using:

$$E = V \cdot Ah$$

and theoretical runtime can be estimated using:

$$t = \frac{E}{P}$$

where E is energy in watt-hours, V is voltage, Ah is capacity, t is runtime, and P is electrical power. These formulas give ideal estimates, while real runtime is affected by battery cut-off voltage, internal resistance, motor load, and losses in the control circuit (National Renewable Energy Laboratory, 2019).

For the low-power mode, motor speed is controlled using pulse width modulation (PWM). PWM changes the average power supplied to the motor by rapidly switching the voltage on and off. This is more efficient than using resistors for speed control, since less energy is wasted as heat (Electronics Tutorials, n.d.).

2.4 Cross-Domain Analogies

The chosen cross-domain analogy is the hair dryer, since both products are compact handheld devices that move air through an internal motor-fan system. Although a hair dryer blows air out and a vacuum cleaner draws air in, both depend on efficient airflow through small ducts.

Li et al. (2025) show that impeller design, guide vanes, and flow separation affect airflow rate, pressure, efficiency, and noise in hair dryers. Their optimized blade and guide-vane design increased airflow by 7.39% and reduced noise by 2.88%. This suggests that the vacuum design should use smooth ducts, limited sharp transitions, and careful motor-fan placement. This is also supported by Ai and Mak (2013), who explain that duct pressure losses are caused by friction and changes in flow direction or cross-sectional area.

2.5 Standards and Technical References

The design is guided by relevant safety and regulatory references for consumer electrical products. IEC 60335-2-2 is relevant for vacuum cleaner safety, including protection against electric shock, overheating, and mechanical hazards (IECEE, 2019). Since the product uses a lithium-ion battery pack, IEC 62133 and UN 38.3 are relevant for rechargeable battery safety and transport testing (Intertek, n.d.-a; Intertek, n.d.-b).

The product would also need to support CE marking requirements before sale in the European/EEA market, including electrical safety and electromagnetic compatibility (European Commission, n.d.). EU Ecodesign requirements for applicable vacuum cleaners are used as benchmark values for motor power, energy use, and noise, although this project does not verify full regulatory compliance (European Commission, 2013). Filtration performance is guided by HEPA/EN 1822 principles, while Regulation (EU) 2023/1542 is relevant for battery traceability and end-of-life handling (European Committee for Standardization, 2009; European Parliament and Council, 2023).

3. Concept Development and Selection

3.1 Ideation Technique

Brainstorming was selected as the primary ideation technique for the concept generation phase. The team conducted several group sessions in which ideas were generated freely without immediate evaluation, followed by clustering and refinement of the most promising directions. This approach was chosen because it encourages creative thinking, leverages the different backgrounds within the team, and produces a broad range of candidate concepts that can later be filtered through a structured selection process.

The brainstorming sessions focused on three main aspects of the product: the overall form factor and ergonomics, the power and charging architecture, and the filtration and dust-collection method. Each team member first generated ideas individually for a fixed period, after which the ideas were presented, grouped, and developed further in plenary discussion. Three main concepts were taken forward from these sessions.

Concept A — Compact integrated handheld (selected concept). A single-piece handheld unit with an internal 4S2P lithium-ion battery, a brushed DC motor with an integrated impeller, a HEPA filter, a transparent dust bin, and USB-C charging. The form factor is similar to commercial products such as the Black+Decker Dustbuster series. This concept prioritises simplicity, low cost, and ease of use, and aligns directly with the priority specifications in Section 1.2.

Concept B — Modular handheld with swappable battery. A handheld unit with an externally removable battery pack that can be exchanged or charged separately from the main body. This concept allows for extended total runtime through spare batteries and easier servicing, but it increases mechanical complexity (electrical contacts, latch mechanism, additional sealing) and overall unit cost. The motor, dust bin, and filter are otherwise similar to Concept A.

Concept C — Pistol-grip with modular attachments. A pistol-grip ergonomic design with a tool-free swappable nozzle system, including a crevice tool, a brush head, and a soft brush. The battery is integrated, but the unit is shipped with multiple attachments for different cleaning scenarios. This concept emphasises versatility and adaptability across

surfaces such as upholstery, car interiors, and keyboards, but it introduces additional cost in the form of multiple accessories and a storage requirement.

3.2 Concept Selection Process

The Pugh selection matrix was used to systematically compare the candidate concepts and the most important component alternatives. The Pugh method evaluates each alternative against a baseline (datum) for each criterion, scoring it as better (+1), worse (-1), or equivalent (0). Each criterion is weighted according to its importance as established in the priority list in Section 1.2 (Product Specifications). The weighted sum determines the preferred alternative. The same approach is applied at three levels: the overall concept, the charging interface, and the battery configuration.

3.2.1 Overall concept selection

Concept A was selected as the datum because it represents the most common form factor for handheld vacuums and aligns most directly with the project brief. Concepts B and C were scored relative to Concept A using the weighted criteria below.

Criterion	Weight	Concept A (datum)	Concept B	Concept C
Usability and convenience	5	0	-1	0
Cost-effectiveness	5	0	-1	-1
Compact form factor	4	0	-1	-1
Battery performance	4	0	+1	0
Cleaning versatility	3	0	0	+1
Manufacturing complexity	3	0	-1	-1
Reliability and sealing	2	0	-1	0
Weighted Total		**0**	**-12**	**-7**

Concept A received the highest weighted score. Concept B's advantage in extended runtime through a spare battery did not outweigh the cost increase, the larger form factor, and the additional sealing required around the removable pack — particularly given that the runtime target of 20 minutes in normal mode is already met by Concept A (Section 4.2.2). Concept C offers improved cleaning versatility, but the additional attachments and the larger packaging conflict with the cost-effectiveness and compact form factor objectives. Concept A was therefore taken forward into detailed design.

3.2.2 Charging interface selection

The charging interface was evaluated separately because it is an important user-facing component. USB-C Power Delivery (PD) was compared with standard USB-C and a traditional DC barrel jack using a Pugh matrix, shown in Appendix [8]. The DC barrel jack was used as the datum because it represents a conventional charging solution for handheld vacuums.

USB-C PD was selected because it combines the accessibility of common USB-C cables with faster charging than standard USB-C. Based on the charging estimate in Section 4.2.4, USB-C PD can reduce charging time to approximately 2–2.5 hours, compared with around 5–6 hours for standard USB charging. Although the PD module has a slightly higher cost, the improved charging convenience supports the customer analysis in Section 2.2, where 41% of consumers indicated willingness to pay more for products that simplify everyday tasks (GfK, 2024).

3.2.3 Battery configuration selection

The battery configuration was evaluated separately because it strongly affects runtime, current delivery, weight, and internal packaging. A 4S2P lithium-ion pack was compared with a removable battery pack, a single Li-Po cell with a boost converter, and a 12-cell NiMH pack using a Pugh matrix, shown in Appendix [9]. The 4S2P lithium-ion pack was used as the datum because it matched the intended final design.

The integrated 4S2P lithium-ion configuration was selected because it matches the 14.4 V motor voltage, provides sufficient current capacity for the estimated 8–10 A motor load, and offers good energy density for a compact handheld product. The single Li-Po with boost converter was rejected due to added electrical losses, cost, and thermal load. The NiMH option was rejected because the number of cells required would increase weight and size. The removable pack was the closest alternative, but the added latch, contacts, and sealing would increase cost and mechanical complexity without improving the core cleaning function. Therefore, the integrated 4S2P pack was carried forward into detailed component selection in Section 4.2.2.

4. Detailed Design

4.1 Design Schematic

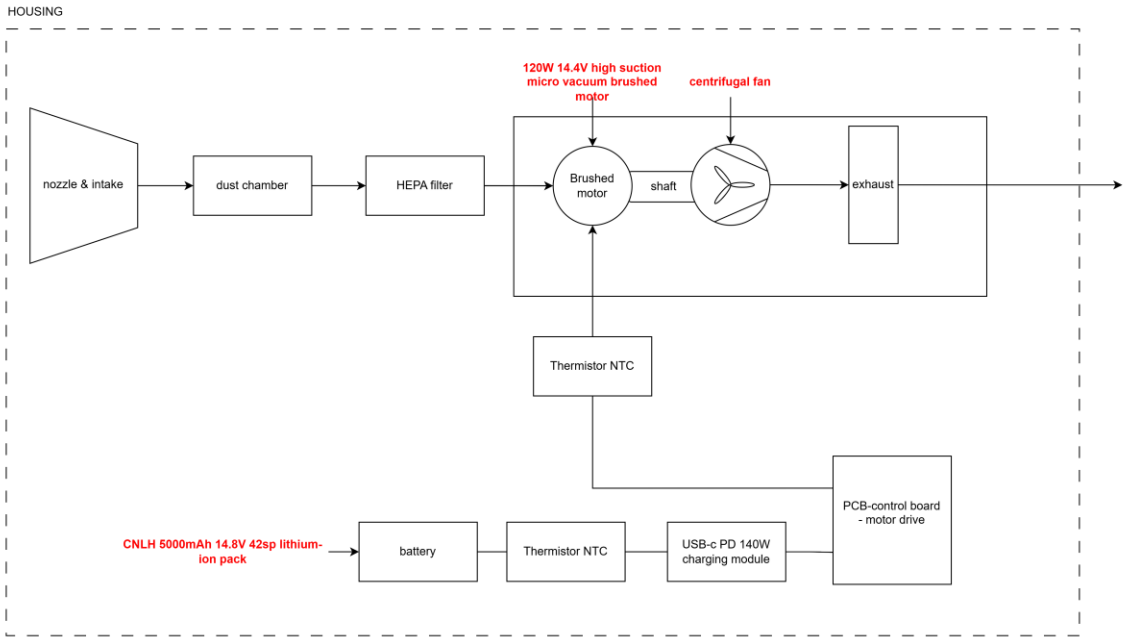


Figure 2: *Design Schematic*

Air enters through the nozzle, which has an opening of 42 × 12.31 mm. The air and dust then enter the dust bin, where the sudden increase in area reduces air velocity and allows larger particles to settle. The dust bin is removable through a latch mechanism, while finer particles are captured by a replaceable HEPA filter.

Behind the filter, the integrated motor-fan unit creates the pressure difference required for suction. A seal between the filter and fan housing reduces leakage, and exhaust air passes around the motor before leaving through the outlet vents, contributing to motor cooling.

The rear section contains the battery pack, PCB control board, and USB-C charging module. The handle includes an on/off switch and a speed-control switch. In low-power mode, the motor is driven through the PWM circuit, while high-power mode bypasses the PWM circuit. The battery pack includes a PCM for electrical protection, while thermistors monitor battery and motor temperature.

4.2 Choice of Standard Components

4.2.1 Motor Selection

The selected motor is a 14.4 V DC vacuum cleaner motor with an integrated fan and a rated power of approximately 120 W (AliExpress, n.d.-a). The motor–fan assembly is designed as a single unit, where the impeller is directly mounted on the motor shaft. This configuration is commonly used in compact vacuum cleaners, as it simplifies the

mechanical design and reduces the need for additional components. The dimensions are shown in Appendix [14].

The integrated fan design allows the motor to be positioned directly in the airflow path, improving overall system efficiency while also contributing to cooling during operation. This reduces thermal stress and supports stable performance, particularly when operating with a HEPA filter, which increases airflow resistance.

The operating current is estimated using:

$$I = \frac{P}{V} = \frac{120}{14{,}4} = 8{,}3\ A$$

This represents the expected current under load. Higher transient currents may occur during startup and have been considered in the selection of the battery and electrical components.

From a regulatory perspective, the selected 120 W motor is well below the 900 W motor-power benchmark used in EU Ecodesign requirements for applicable vacuum cleaners (European Commission, 2013). This supports the design goal of low energy use. Since no full prototype is manufactured in this project, noise and thermal behaviour are treated as design considerations rather than verified performance values.

The selection of an integrated motor–fan unit supports the design objective of cost-effectiveness and simplicity, as it reduces manufacturing complexity and component count.

4.2.1.1 Choice of Brushed Motor:

A brushed DC motor was selected instead of a brushless DC (BLDC) motor due to its lower cost and simpler control requirements. Unlike BLDC motors, which require electronic controllers for commutation, brushed motors can operate directly from a DC power supply. This eliminates the need for additional control electronics and reduces both system complexity and cost (Equipmake, n.d.; AKT Motor & Drive, n.d.).

While brushed motors generally have lower efficiency and shorter lifespan compared to brushless alternatives, these disadvantages are acceptable for the intended application. The design brief prioritizes affordability, simplicity, and ease of use over high-end performance and durability.

The brushed motor therefore provides a suitable balance between performance, cost, and implementation complexity, making it well aligned with the overall design goals.

4.2.1.2 Suction estimation:

The suction performance is estimated based on the relationship between air power, pressure difference, and airflow:

$$P = \Delta P \cdot Q$$

Assuming a brushed DC motor efficiency of approximately 70%, which is within the typical operating range of 65–80% (LeanMotor, n.d.), the mechanical shaft power is:

$$P_{shaft} = 0{,}7 \cdot 120 = 84W$$

With an estimated airflow of:

$$Q \approx 0{,}015\, m^3/s$$

corresponding to approximately 15 L/s, the value is based on published specifications for a comparable handheld vacuum cleaner with a similar power rating and application (Scheppach, n.d.). As the motor manufacturer does not provide airflow data, this reference is used to establish a representative operating estimate. The corresponding pressure difference is calculated as:

$$\Delta P = \frac{84}{0{,}015} \approx 5600\, Pa$$

This represents an **idealised upper-bound estimate**, assuming that all mechanical shaft power is converted into useful airflow.

In practice, suction performance is reduced by losses in the filter, air sealing, and internal airflow path. Studies indicate that up to one-third of vacuum power can be lost due to such inefficiencies (Lin et al., 2024).

Applying a one-third loss factor, the effective air power becomes:

$$P_{air} = \frac{2}{3} \cdot 84 = 56W$$

Then:

$$\Delta P = \frac{56}{0{,}015} \approx 3730\, Pa$$

Based on these calculations, the suction pressure is estimated to lie between an idealised upper bound of approximately **5600 Pa** and a loss-adjusted estimate of approximately **3700 Pa**.

Due to uncertainties in airflow, efficiency, and additional pressure losses from components such as the filter and internal ducts, the actual performance is expected to fall within approximately **3700–5600 Pa**, which is consistent with typical handheld vacuum cleaners operating in the range of 3000–6000 Pa (Dreame, n.d.).

4.2.2 Battery Selection

The selected battery is a lithium-ion battery pack with a configuration of 4S2P, a nominal voltage of 14.8 V, and a capacity of 5000 mAh (AS Battery, n.d.). The battery has dimensions that make it suitable for integration within the main body of the handheld vacuum cleaner. The dimensions are shown in Appendix [12].

The 4S configuration was selected to match the nominal voltage of the motor (14.4 V), ensuring efficient operation without the need for additional voltage conversion. The 2P configuration increases both the total capacity and current delivery capability of the battery. This is necessary to support the motor's current demand, estimated at approximately 8–10 A under load. According to the manufacturer specifications, the battery supports a continuous discharge current of up to 15 A, which provides sufficient margin for stable operation.

The total energy stored in the battery is calculated as:

$$E = V \cdot Ah = 14{,}8 \cdot 5 = 74Wh$$

The theoretical runtime of the vacuum cleaner is estimated using:

$$t = \frac{E}{P} = \frac{74}{120} \approx 0{,}62\ h \approx 37\ \text{min}$$

This represents an ideal case. In practical operation, runtime will be lower due to battery cut-off voltage, internal resistance, voltage drop, control circuit losses, and increased motor load caused by airflow resistance and filter restrictions. Brushed DC motors typically operate with efficiencies in the range of approximately 65–80% (LeanMotor, n.d.). Applying this range results in an estimated practical runtime of approximately 22–30 minutes in normal mode.

An eco mode is included by reducing motor power through PWM. This increases the estimated runtime to approximately 40–55 minutes and is useful for lighter cleaning tasks where full suction is not required, such as cleaning fabrics, upholstery, or desks. Since the motor operates at lower power in this mode, energy consumption and

component stress are also reduced, supporting the design goals of usability and energy efficiency.

These values are consistent with performance observed in commercially available cordless vacuum cleaners. Independent testing of over 100 cordless vacuums shows that runtime in normal or high-power modes typically ranges between approximately 15–30 minutes, while longer runtimes are often achieved in lower-power eco modes (Vacuum Wars, 2020).

Overall, the selected battery provides a suitable balance between runtime, power delivery, size, and cost, aligning well with the design objective of a practical and user-friendly handheld vacuum cleaner.

4.2.2.1 Battery Protection (PCM)

The selected battery pack includes an integrated Protection Circuit Module (PCM), as specified by the manufacturer (AS Battery, n.d.). The PCM provides essential safety functions, including protection against overcharging, over-discharging, overcurrent, and short circuits, as well as cell balancing.

These functions are important because lithium-ion cells can become unsafe if they operate outside their allowed voltage or current limits. The PCM therefore supports the battery safety requirements discussed in Section 2.5, especially the principles behind IEC 62133 for portable rechargeable batteries.

Compared to a full Battery Management System (BMS), which includes more advanced monitoring and communication features, a PCM provides basic protection. For this application, a PCM is considered sufficient because the design does not require advanced battery monitoring or control functionality (Jetray Battery, n.d.).

4.2.3 Temperature Monitoring Component Selection

To ensure safe operation of the lithium-ion battery during charging, a temperature monitoring component is included in the design. A 10 kΩ NTC (Negative Temperature Coefficient) thermistor (Vishay NTCLE300E3103SB) was selected for this purpose (Farnell, n.d.).

NTC thermistors decrease in resistance with increasing temperature, enabling real-time monitoring of temperature. The selected thermistor operates over a wide temperature range (−40°C to 125°C), ensuring reliable measurement under expected conditions.

In lithium-ion battery systems, safe charging is typically limited to a narrower temperature range, approximately 0°C to 45–50°C, to prevent degradation and ensure safe operation (Texas Instruments, n.d.). The thermistor placed in thermal contact with the battery pack is connected to the charging module, allowing it to reduce or disable charging if these limits are exceeded.

In addition, a second NTC thermistor is used to monitor the temperature of the motor during operation. This thermistor is connected to the control system, enabling the vacuum to reduce power or shut down if excessive temperatures are detected.

Together, these temperature monitoring components provide protection during both charging and operation, improving overall system safety and reliability beyond the basic electrical protection provided by the PCM.

4.2.4 Charging Module Selection

The selected charging module is a USB Type-C Power Delivery (PD) charger/discharger module compatible with 4S lithium-ion battery packs (AliExpress, n.d.-b). The module provides a charging voltage of 16.8 V, matching the required charging voltage for the selected battery.

USB-C Power Delivery (PD) enables higher input voltages (up to 20 V) and increased power transfer compared to standard USB charging, allowing significantly faster charging (USB Implementers Forum, n.d.). This aligns with the project goal of ease of use, as it reduces downtime between uses.

The charging time can be estimated using:

$$t = \frac{C}{I}$$

Where the battery capacity is $C = 5Ah$.

For a standard USB input (≈1 A), the charging time is approximately 5–6 hours. With USB-C PD, a higher charging current of approximately 2.5–3 A can be achieved, resulting in a charging time of approximately **2–2.5 hours** when accounting for charging losses and the constant current/constant voltage (CC/CV) charging profile (Battery University, n.d.).

This represents a reduction in charging time of more than 50% compared to conventional USB charging. For comparison, typical handheld vacuum cleaners often have charging times in the range of approximately 3.5–5 hours (JustAnswer, n.d.), indicating that the selected solution provides a competitive advantage in terms of charging speed.

In addition, the use of USB-C enhances user-friendliness and flexibility. If the original charger is unavailable, the device can still be charged using standard USB-C cables, which are widely available and commonly used for consumer electronics. This aligns with the project brief, where ease of use and accessibility are key design considerations. Furthermore, as identified in the customer analysis (Section 2.2), users place a strong

emphasis on convenience and time-saving features, with 41% of consumers willing to pay more for products that simplify everyday tasks (GfK, 2024, p. 16). This supports the inclusion of a faster and more accessible charging solution despite the slightly higher component cost.

The module is placed near the external charging port to simplify integration and improve heat dissipation. Additionally, the module supports temperature feedback (NTC), enabling safe charging in combination with the selected thermistor.

Although the module includes bidirectional functionality, this feature is not essential for the application but does not negatively impact system performance.

4.2.5 Electrical Circuit Selection

A two-speed motor control circuit was selected to provide one high-power mode for normal cleaning and one lower-power mode for more sensitive surfaces, such as fabrics, leather seats, or desks with loose papers. A resistor-based low-speed circuit was first considered, but this was rejected because the excess electrical energy would be converted into heat, reducing efficiency and wasting battery capacity. A microcontroller-based solution, such as an ESP32, was also considered, but was judged unnecessarily complex and costly for a simple handheld vacuum cleaner.

After further research and consultation with members of the student organization **The Resistance**, a PWM-based solution was selected for the lower-power mode. PWM controls the average motor power by rapidly switching the supply voltage on and off. By changing the duty cycle, the motor output can be reduced without wasting as much energy as a resistor-based solution. A 555 timer PWM circuit [2], using a 555 timer IC [1], was selected and modified to include an on/off switch and a speed-control switch. The speed-control switch either routes the motor through the PWM circuit or bypasses it for full-power operation. This became the final circuit-board schematic [3].

The final parts list is as follows:

Capacitors	Resistors	Diodes	IC	MOSFET	Potmeter	PCB
1nF, 100nF	1kΩ, 5kΩ	1N4001 x 2	NE555	IRFZ44N N-channel	10kΩ	Custom order

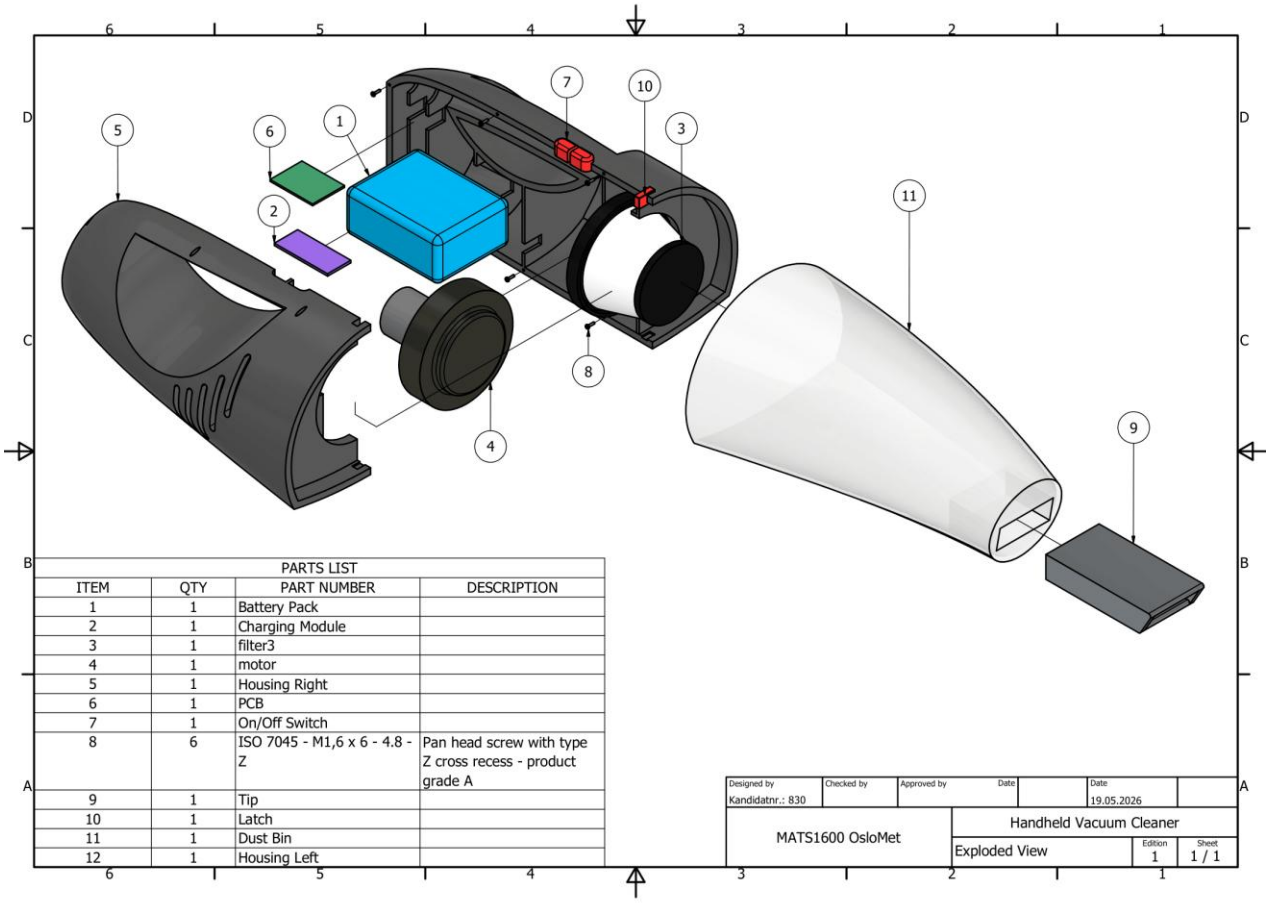
4.2.6 Filter Selection

A replaceable HEPA filter was selected to capture fine dust particles and improve exhaust air quality. The selected filter is designed for compact handheld vacuum cleaners, making it suitable for integration within the proposed design (Clas Ohlson, n.d.). The dimensions are shown in Appendix [13].

HEPA filters are designed for high filtration efficiency, typically capturing very small particles around 0.3 µm (Donaldson Company, n.d.). The main drawback is increased

airflow resistance, which was considered during motor selection and contributed to the choice of a 120 W motor. A replaceable filter also supports usability, since the user can maintain filtration performance over time. The filter choice supports the filtration requirements discussed in Section 2.5, including EN 1822 principles.

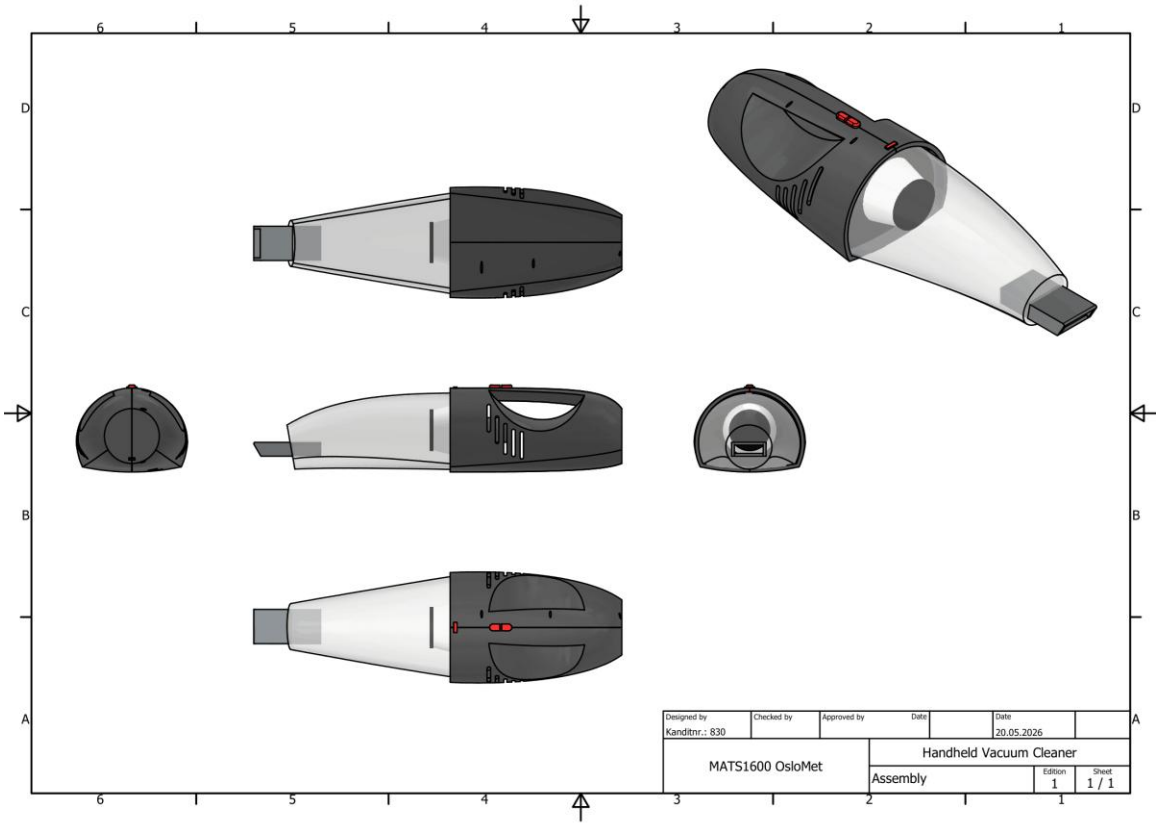
4.3 Technical Design of Unique Components



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1	1	Battery Pack	
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6	1	PCB	
7	1	On/Off Switch	
8	6	ISO 7045 - M1,6 x 6 - 4.8 - Z	Pan head screw with type Z cross recess - product grade A
9	1	Tip	
10	1	Latch	
11	1	Dust Bin	
12	1	Housing Left	

Designed by	Checked by	Approved by	Date		Date
Kandidatnr.: 830					19.05.2026

MATS1600 OsloMet | Handheld Vacuum Cleaner | Exploded View | Edition 1 | Sheet 1 / 1



Component	Recommended material	Reason
Main casing	ABS	Cheap, easy to mould and commonly used for appliance housings. Good enough for a practical low-cost vacuum cleaner.
Dust bin	Clear PC	Keeps the bin transparent and is tougher than acrylic, so it is less likely to crack if the unit is knocked or dropped.
Buttons	ABS	Low cost, easy to mould and suitable for simple button and trigger parts.
Internal button/latch mechanisms	POM	Used only for the small moving parts where smoother movement and better wear resistance are needed.
Dust bin latch	POM	A practical choice for repeated opening and closing without making the whole product more expensive.
Seals	TPE	Lower-cost sealing material that is widely used in consumer products and works well enough for dust sealing.
Grip areas	TPE overmould	Used only in limited hand-contact areas to improve grip without adding too much cost.
Internal dust guides / ducts	PP	Light, cheap and suitable for simple internal plastic parts that are not highly loaded.

Overall, the material choices are based on keeping the product affordable, easy to manufacture, and durable enough for normal household use. Dimensioned 2D drawings of the vacuum assembly and main components are included in Appendix [10]–[18], showing how the casing, dust bin, motor, battery, filter, and internal parts fit together within the custom housing.

4.4 Manufacturing Plan

To balance Norway's high labor costs with the need for high-quality output, this project utilizes a semi-automated modular assembly strategy. The focus is on local high-value integration and strict adherence to EEA (European Economic Area) safety standards.

Component	Source Category	Specification / Standard
DC Motor & Li-ion Cells	International Procurement	RoHS & REACH Compliant
PCB (Charging/PCM)	Pre-fabricated PCB with SMT-assembled components	IPC-A-610 Class 2
Casing	China	Injection-Moulded ABS
HEPA Filter	Local Retail/Wholesale	EN 1822 (H13 Grade)

Assembly process flow:

1. Electronics sub assembly:
    - Spot-weld Li-Ion batteries into 4S2P configuration
    - Integrate Protection Circuit Module (PCM) and NTC thermistor.
2. Aero-Core integration:
    - Press-fit impeller onto the 14.4 V brushed DC motor (Section 4.2.1)
    - Dynamic-balance the impeller and verify run-out before bonding the second NTC thermistor (Section 4.2.3) to the motor housing.
3. Main Chassis Assembly:
    - Install Aero-core and battery pack to injector molded casing.
    - Connect wiring through snap connectors wherever possible to reduce soldering.
4. Filtration:

- Insert HEPA filter and seal before testing airflow.

The complete manufacturing plan is summarized in Figure 3, which shows the four parallel sub-assemblies (electronics, aero-core, filtration and plastics) feeding the semi-automated main chassis line, followed by four parallel end-of-line quality-control gates and final CE marking and packaging. Sub-assemblies A and B carry the highest-risk operations identified by the FMEA (Section 5.1)—battery spot-welding and motor balancing—and are therefore staffed by trained technicians, while molding and SMT operations are outsourced to qualified international suppliers under RoHS and REACH compliance.

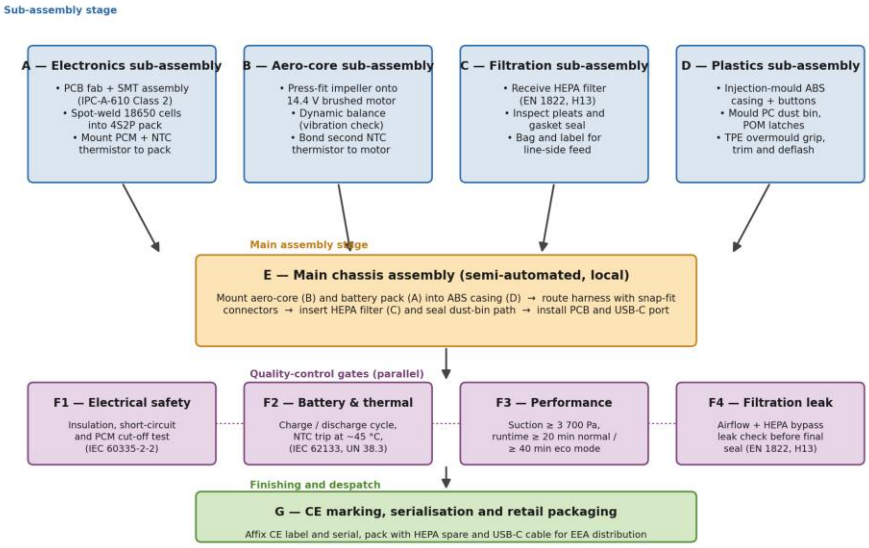


Figure 3: *Manufacturing process flow for the handheld vacuum cleaner, showing parallel sub-assemblies (A–D), the main chassis line (E), parallel end-of-line QC gates (F1–F4), and final CE marking and despatch (G).*

Quality control and end-of-line testing

Every finished unit is tested at four parallel gates before being released for packaging. The gates are configured to verify that the unit meets both the priority specifications listed in Section 1.2 and the safety standards summarized in Section 2.5.

F1 — Electrical safety: insulation resistance, short-circuit response, and verification that the Protection Circuit Module (Section 4.2.2) opens the discharge path at the manufacturer-specified over-current threshold. The test references IEC 60335-2-2 for safe construction of mains- and battery-powered vacuum cleaners (IECEE, 2019).

F2 — Battery and thermal: one complete charge/discharge cycle on the 4S2P 5 Ah pack (Section 4.2.2) through the USB-C PD module (Section 4.2.4), with both NTC thermistors (Section 4.2.3) instrumented. Charging must reduce or interrupt at approximately 45 °C as required by the lithium-ion safe-charging window (Texas Instruments, n.d.). Lot samples are submitted to an external lab for IEC 62133 and UN 38.3 testing (Intertek, n.d.-a; Intertek, n.d.-b).

F3 — Performance: suction pressure measured at the dust-bin inlet against a calibrated orifice; the unit passes if the measured value is at least 3 700 Pa, matching the loss-adjusted lower bound calculated in Section 4.2.1. Runtime is verified on a sample basis at ≥ 20 min in normal mode and ≥ 40 min in eco mode, matching the priority targets in Section 1.2 and the runtime estimate in Section 4.2.2.

F4 — Filtration leak: an aerosol challenge test downstream of the HEPA element with a particle counter to confirm that the seal does not bypass. The pass criterion follows EN 1822 for H13-class filters (European Committee for Standardization, 2009), as specified for the HEPA element in Section 4.2.6.

Compliance, traceability and CE marking

Each PCB carries an IPC-A-610 Class 2 inspection record from the supplier; each battery pack carries a discharge signature linked to its lot number; and every finished unit is laser-engraved with a serial number that ties back to the motor batch, PCB batch, battery batch and HEPA filter batch. The records are retained for ten years to support the CE Declaration of Conformity (European Commission, n.d.), which is required for placement on the EEA market as discussed in Section 2.5. Units that pass all four QC gates proceed to stage G in Figure 3, where the CE label is affixed and the unit is packed with one spare HEPA filter and a standard USB-C cable for retail distribution.

Summary

The semi-automated modular approach keeps the highest-value steps—final assembly, end-of-line testing and certification—local to Norway, while higher-volume operations (cell manufacturing, motor production, PCB fabrication and plastic molding) are sourced from established international suppliers under RoHS and REACH compliance. This split is consistent with the cost-effectiveness objective set out in Section 1.1, the priority targets for runtime, charging time and form factor in Section 1.2, and the material choices in Section 4.3.

5. Validation of Design

5.1 FMEA

A FMEA analysis was conducted to identify and assess potential failure modes in the handheld vacuum, with a focus on reliability, safety, and overall functionality. The analysis includes central components such as battery pack, motor, HEPA-filter, air intake, control board (PCB), and charging module. The risks were assessed using Risk Priority Number (RPN) and were calculated as the product of Severity x Occurrence x Detection. The full FMEA table is shown in Appendix [7].

The results show that the dust chamber and motor shaft had the highest RPN values, mainly because of the risk for reduced suction, overheating, and mechanical failure. HEPA-filter and the air intake were also identified as critical areas because the blockage can reduce air flow and increase the load on the motor. Based on this some measures

and notes as improved sealing, stronger material, better cooling and easier maintenance were considered during the designing phase, to ultimately reduce risk and improve the product`s robustness.

5.2 Testing / Validation

The electrical circuit was first built using a virtual breadbox on TinkerCAD.com. This way we could simulate the circuit to verify that it works as expected. After we built the circuit virtually (appendix [4]) and made sure that it worked, we decided to test it in real life too with an oscilloscope (appendix [5]), as the virtual simulation was sub-optimal. TinkerCAD's simulations aren't meant to simulate fast pulses like that of a PWM. The test with the oscilloscope showed that the design works as intended, and most importantly, in a real life scenario. After we verified the circuit with real components, we decided to order a custom PCB board for the circuit to see how little space it would end up taking [6]. The final dimension of the custom PCB board is 30x40 mm.

6. Evaluation of Final Design

The final design appears technically feasible and is expected to meet the main project objectives. The selected motor, battery, HEPA filter, and USB-C PD charging module provide a practical balance between suction performance, runtime, charging time, safety, and user convenience. The calculated suction pressure is above the initial target of 3000 Pa, and the battery estimate indicates that the design should meet the required runtime in normal and eco mode.

A strength of the design is the use of standard components, such as the motor-fan unit, battery pack, charging module, thermistors, and HEPA filter. This reduces manufacturing complexity, cost, and assembly risk. The PWM circuit was also validated through simulation and physical testing with an oscilloscope, confirming that the speed-control concept works.

The main limitation is that a complete prototype was not manufactured. Therefore, suction performance, airflow losses, noise, thermal behavior, and filtration sealing could not be physically verified. The oscilloscope test also showed voltage spikes in the PWM signal, indicating that additional electrical protection or filtering should be considered before final implementation.

Overall, the design meets the project goals at the design stage, but a full prototype would be needed to confirm real-world performance and identify any remaining issues before production.

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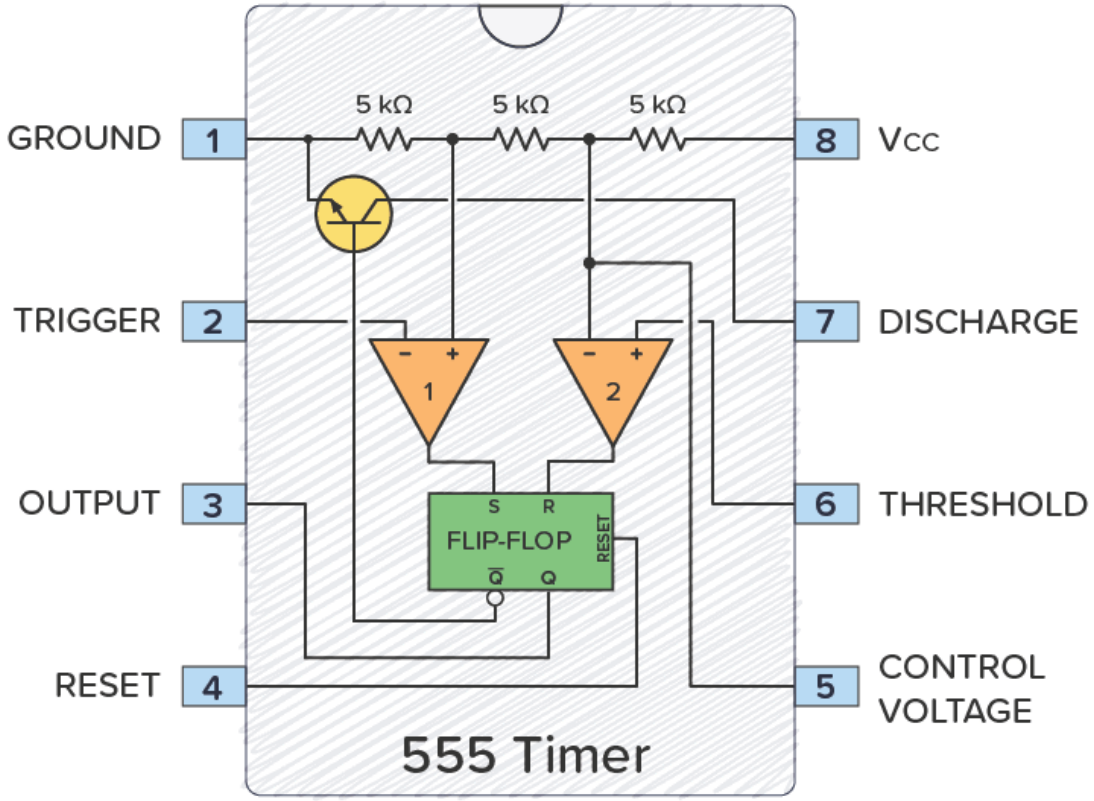
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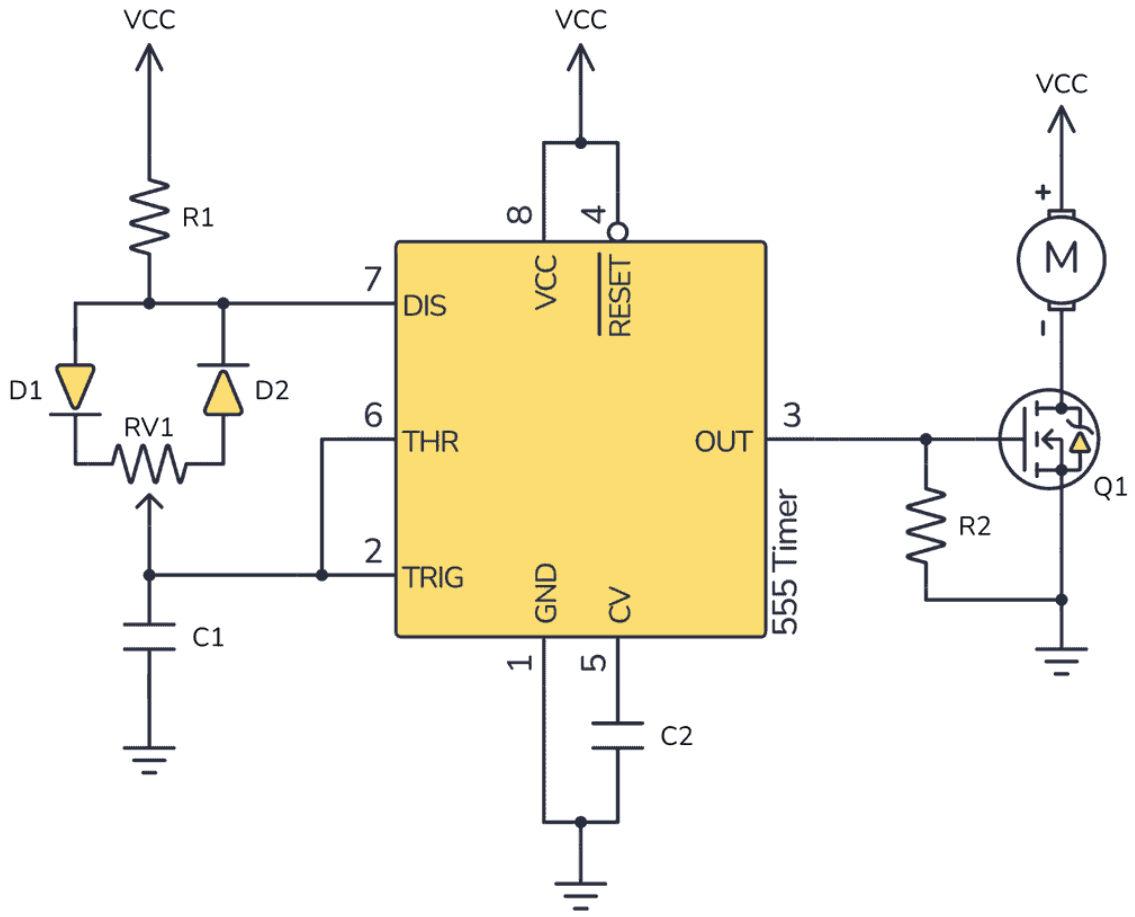
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8. Appendix

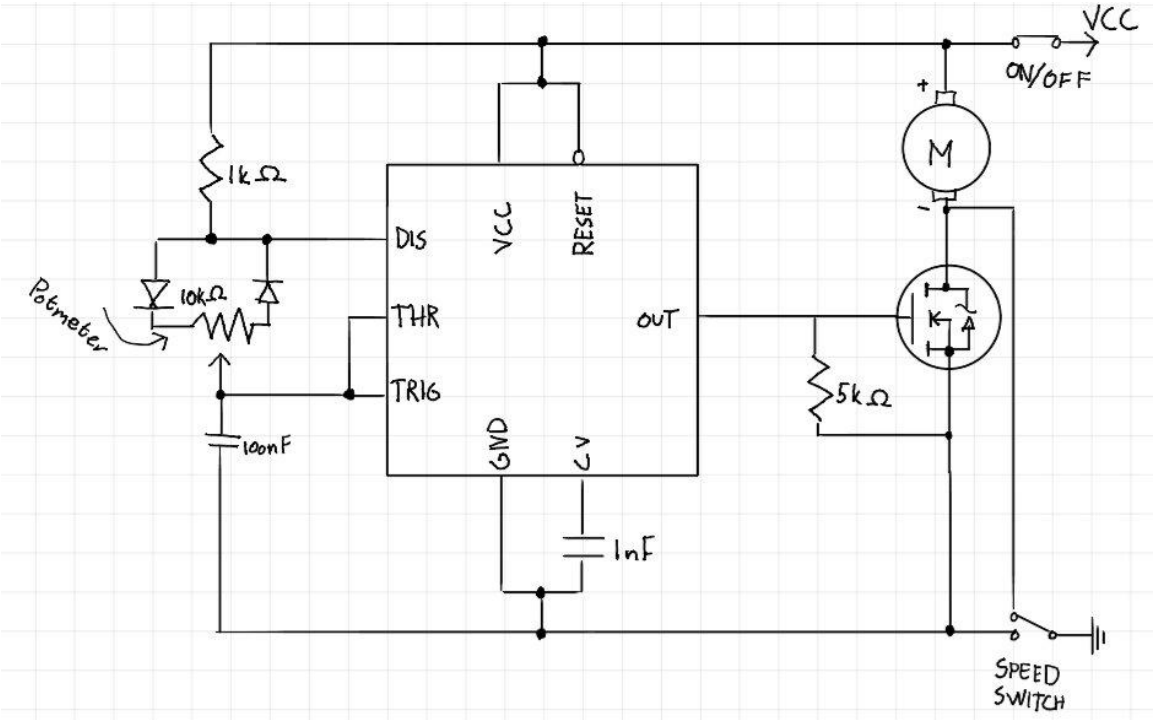
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5 kΩ
5 kΩ
8
Vcc
TRIGGER
2
7
DISCHARGE
1
2
OUTPUT
3
S
R
FLIP-FLOP
RESET
6
THRESHOLD
RESET
4
5
CONTROL
VOLTAGE
555 Timer

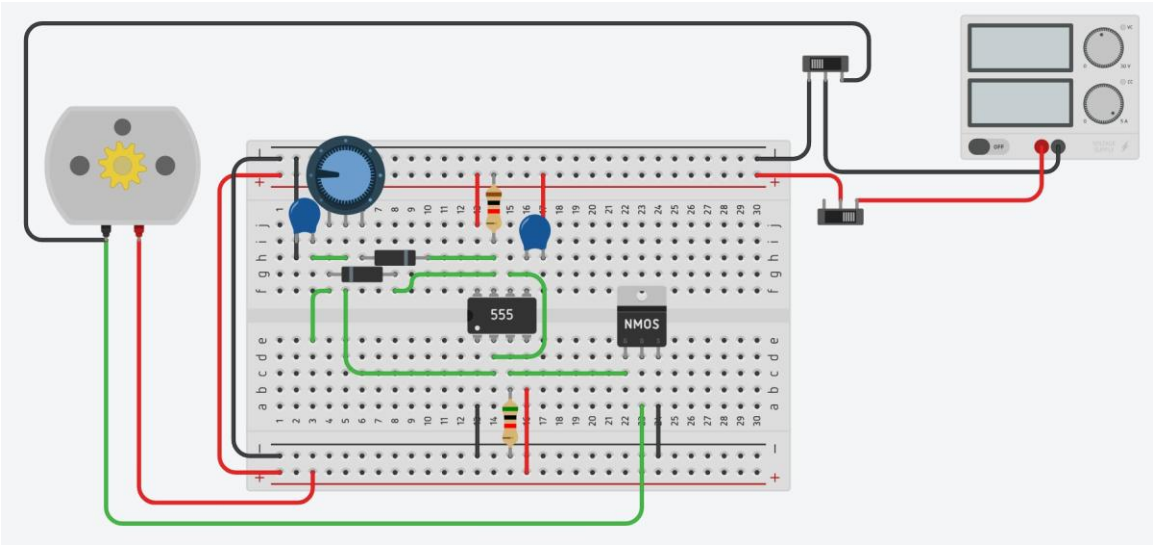
[2]


VCC
VCC
VCC
R1
8
4
7
DIS
VCC
RESET
D1
D2
RV1
6
THR
OUT
3
M
+
-
Q1
R2
2
TRIG
GND
CV
555 Timer
C1
1
5
C2

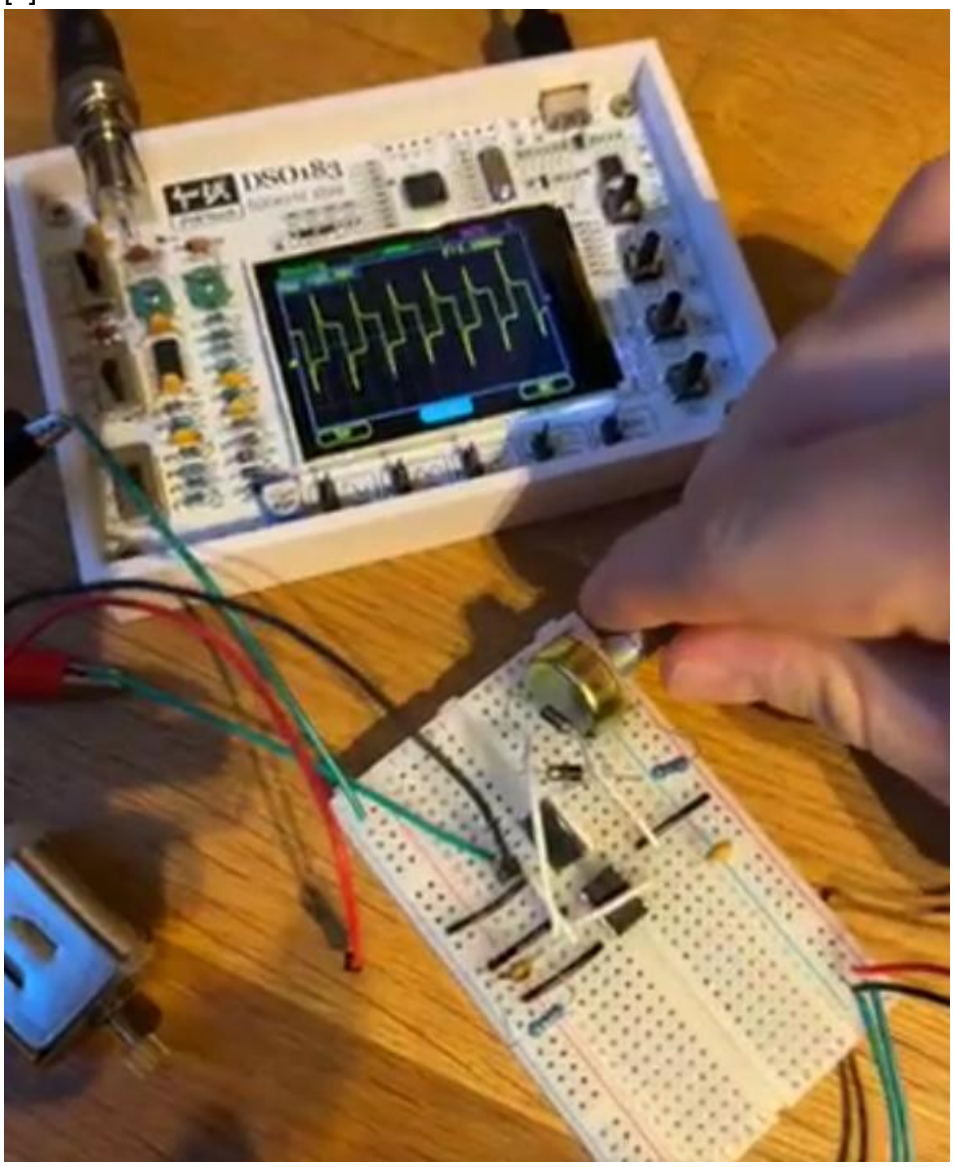
[3]


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M
1kΩ
DIS
VCC
RESET
Potmeter
10kΩ
THR
OUT
K
TRIG
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100nF
GND
CV
1nF
SPEED
SWITCH

[4]


555
NMOS

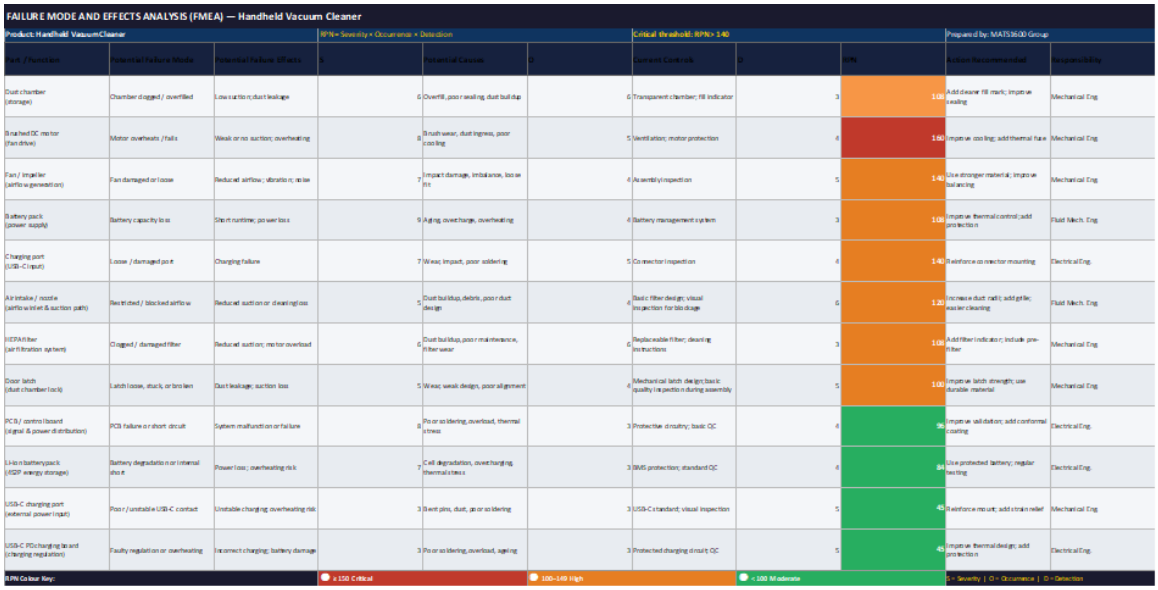
[5]



[6]



[7]



FAILURE MODE AND EFFECTS ANALYSIS (FMEA) — Handheld Vacuum Cleaner

Product: Handheld VacuumCleaner | RPN = Severity × Occurrence × Detection | Critical threshold: RPN > 140 | Prepared by: MATS3600 Group

Part / Function	Potential Failure Mode	Potential Failure Effects	S	Potential Causes	O	Current Controls	D	RPN	Action Recommended	Responsibility
Dust chamber (storage)	Chamber clogged / overfilled	Low suction; dust leakage	6	Overfill, poor sealing, dust build up	6	Transparent chamber; fill indicator	3	108	Add clearer fill mark; improve sealing	Mechanical Eng.
Brushed DC motor (fan drive)	Motor overheats / fails	Weak or no suction; overheating	8	Brush wear, dust ingress, poor cooling	5	Ventilation; motor protection	4	160	Improve cooling; add thermal fuse	Mechanical Eng.
Fan / impeller (airflow generation)	Fan damaged or loose	Reduced airflow; vibration; noise	7	Impact damage, imbalance, loose fit	4	Assembly inspection	5	140	Use stronger material; improve balancing	Mechanical Eng.
Battery pack (power supply)	Battery capacity loss	Short runtime; power loss	9	Aging, overcharge, overheating	4	Battery management system	3	108	Improve thermal control; add protection	Fluid Mech. Eng.
Charging port (USB-C input)	Loose / damaged port	Charging failure	7	Wear, impact, poor soldering	5	Connector inspection	4	140	Reinforce connector mounting	Electrical Eng.
Air intake / nozzle (airflow inlet & suction path)	Restricted / blocked airflow	Reduced suction or cleaning loss	5	Dust buildup, debris, poor duct design	4	Basic filter design; visual inspection for blockage	6	120	Increase duct radii; add grille; easier cleaning	Fluid Mech. Eng.
HEPA filter (air filtration system)	Clogged / damaged filter	Reduced suction; motor overload	6	Dust buildup, poor maintenance, filter wear	6	Replaceable filter; cleaning instructions	3	108	Add filter indicator; include pre-filter	Mechanical Eng.
Door latch (dust chamber lock)	Latch loose, stuck, or broken	Dust leakage; suction loss	5	Wear, weak design, poor alignment	4	Mechanical latch design; basic quality inspection during assembly	5	100	Improve latch strength; use durable material	Mechanical Eng.
PCB / control board (signal & power distribution)	PCB failure or short circuit	System malfunction or failure	8	Poor soldering, overload, thermal stress	3	Protective circuitry; basic QC	4	96	Improve validation; add conformal coating	Electrical Eng.
Li-ion battery pack (MCP energy storage)	Battery degradation or internal short	Power loss; overheating risk	7	Cell degradation, overcharging, thermal stress	3	BMS protection; standard QC	4	84	Use protected battery; regular testing	Electrical Eng.
USB-C charging port (external power input)	Poor / unstable USB-C contact	Unstable charging; overheating risk	3	Bent pins, dust, poor soldering	3	USB-C standard; visual inspection	5	45	Reinforce mount; add strain relief	Mechanical Eng.
USB-C PD charging board (charging regulation)	Faulty regulation or overheating	Incorrect charging; battery damage	3	Poor soldering, overload, ageing	3	Protected charging circuit; QC	5	45	Improve thermal design; add protection	Electrical Eng.

RPN Colour Key: ● ≥ 150 Critical ● 100–149 High ● < 100 Moderate | S = Severity | O = Occurrence | D = Detection

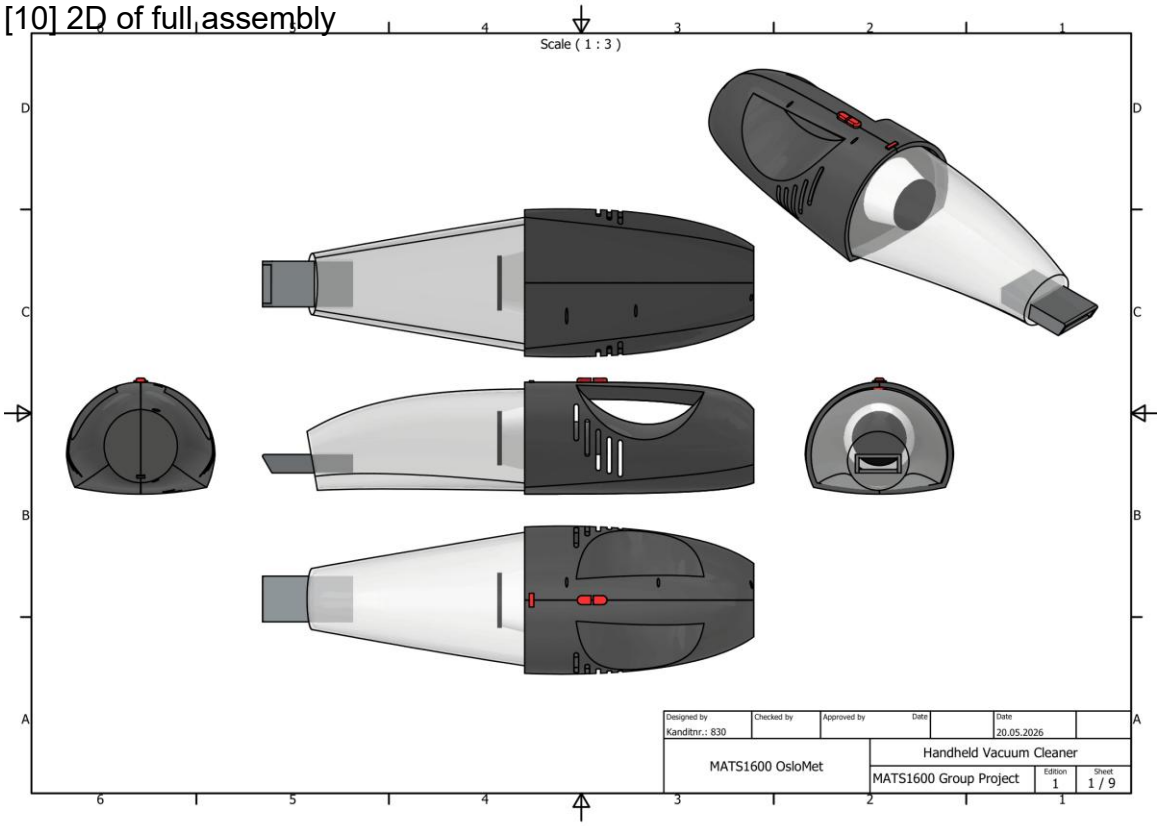
[8]

Criterion	Weight	DC Barrel (datum)	Standard USB-C	USB-C PD
Charging speed	5	0	-1	+1
User accessibility (cable availability)	5	0	+1	+1
Cost of charging module	3	0	0	-1
Mechanical robustness of connector	3	0	+1	+1
Compliance with USB-IF standards	2	0	+1	+1
Future-proofing	2	0	+1	+1
Compatibility with existing user chargers	4	0	+1	+1
Weighted Total		**0**	**+14**	**+17**

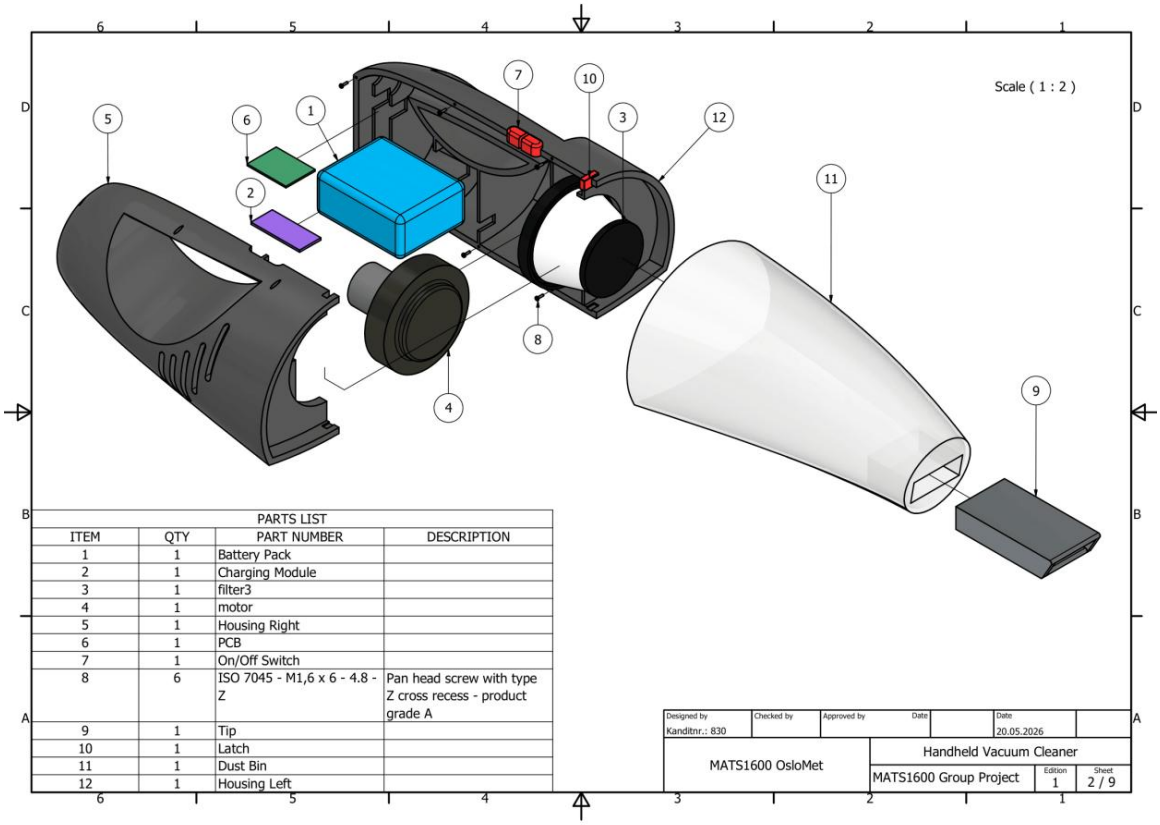
[9]

Criterion	Weight	4S2P Li-ion (datum)	Removable pack	Single Li-Po + boost	NiMH 12-cell
Cost	5	0	-1	-1	+1
Energy density (size/weight)	5	0	-1	-1	-1
Continuous discharge ≥ 10 A	5	0	0	-1	-1
Compact form factor	4	0	-1	0	-1
Runtime ≥ 20 min normal mode	4	0	0	-1	-1
Manufacturing complexity	3	0	-1	-1	0
Voltage match to motor (14.4 V)	3	0	0	-1	-1
Cycle life	2	0	0	0	+1
Field replaceability	2	0	+1	0	0
Weighted Total		**0**	**-10**	**-22**	**-15**

[10] 2D of full assembly



[11] Exploded view



PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	Battery Pack	
2	1	Charging Module	
3	1	filter3	
4	1	motor	
5	1	Housing Right	
6	1	PCB	
7	1	On/Off Switch	
8	6	ISO 7045 - M1,6 x 6 - 4.8 - Z	Pan head screw with type Z cross recess - product grade A
9	1	Tip	
10	1	Latch	
11	1	Dust Bin	
12	1	Housing Left	

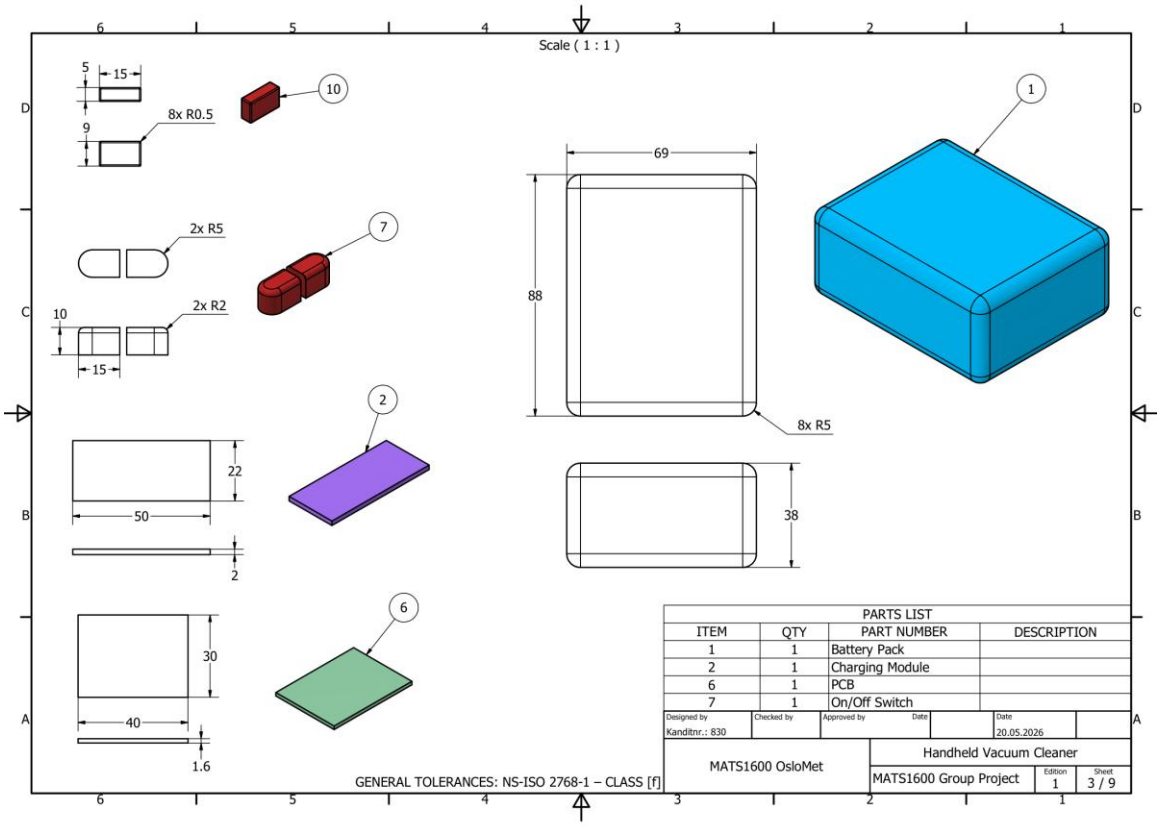
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Kanditnr.: 830				20.05.2026

MATS1600 OsloMet

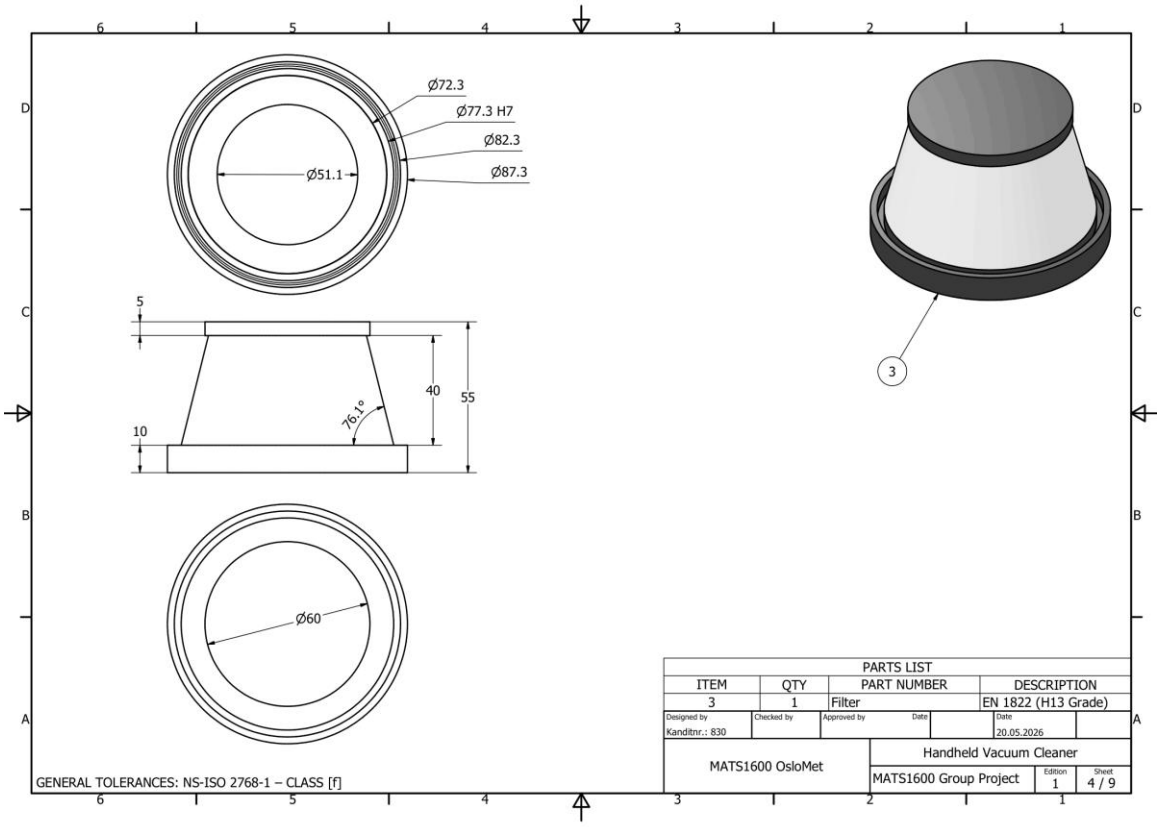
Handheld Vacuum Cleaner

MATS1600 Group Project | Edition 1 | Sheet 2 / 9

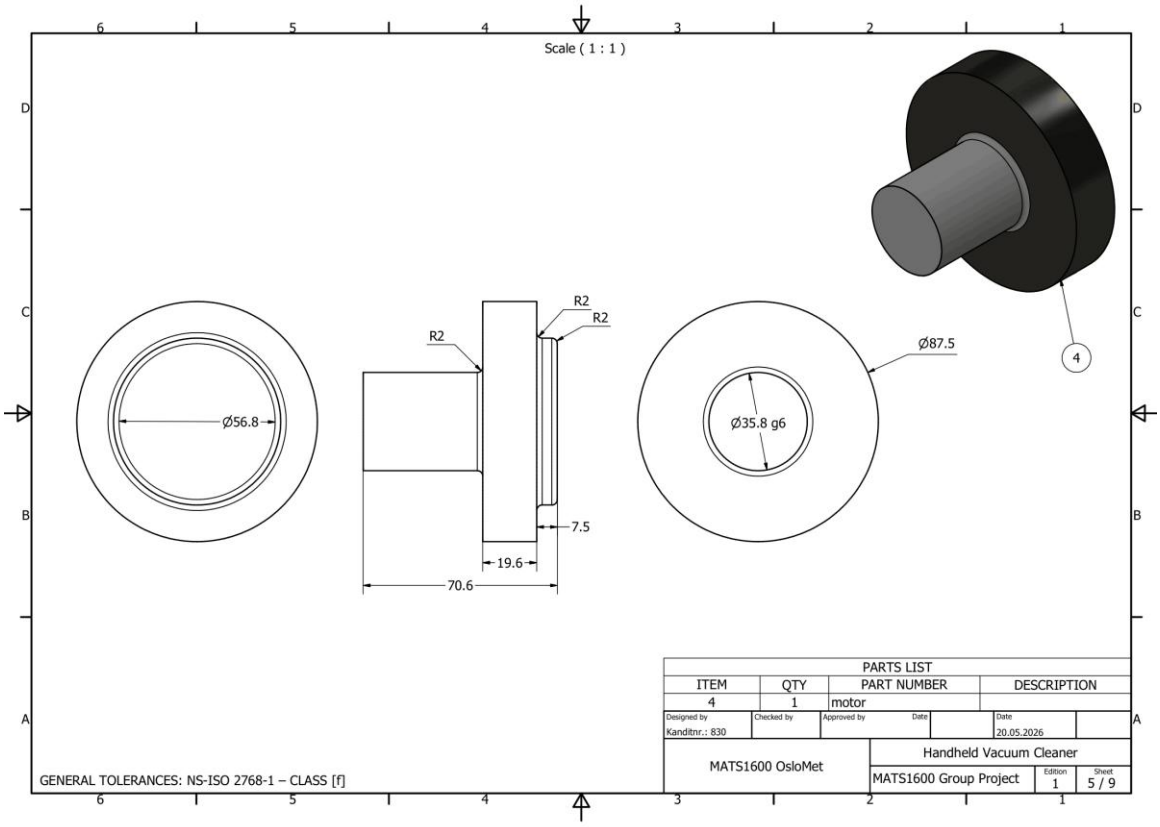
[12] 2D of battery, charging module, PCB and switch



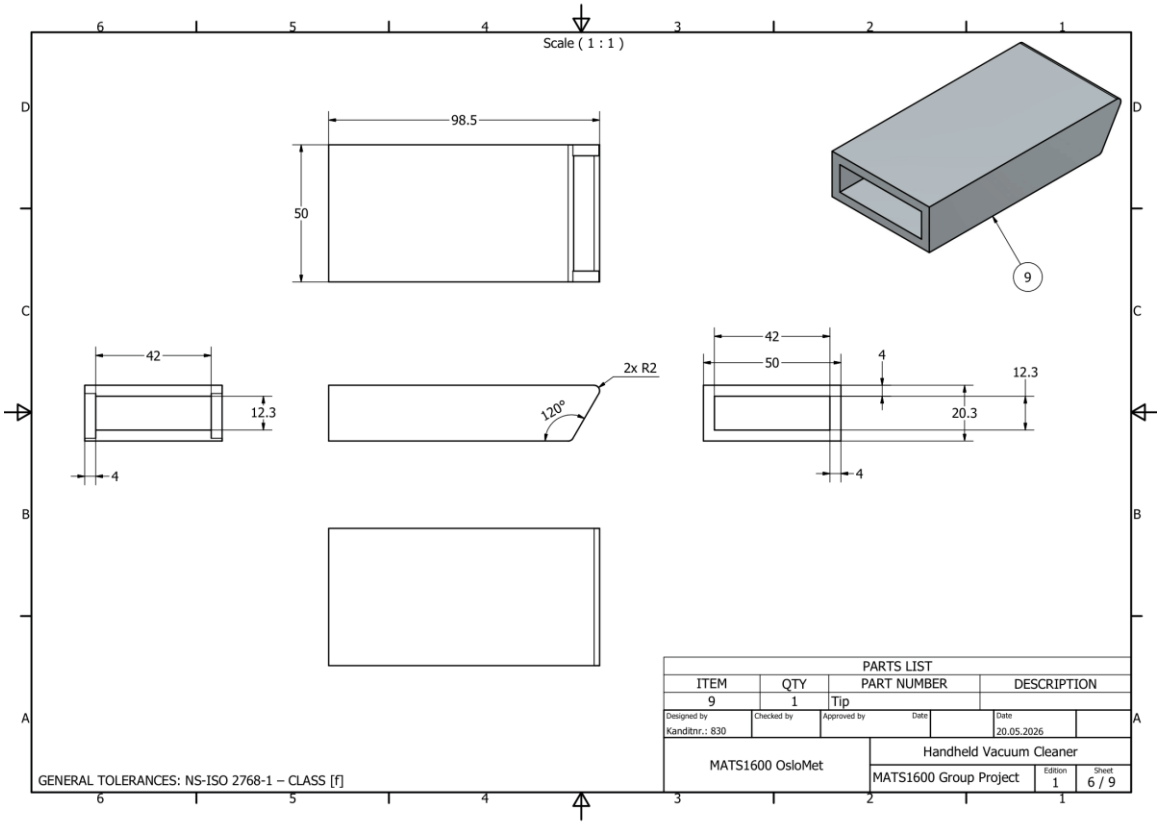
[13] 2D of filter



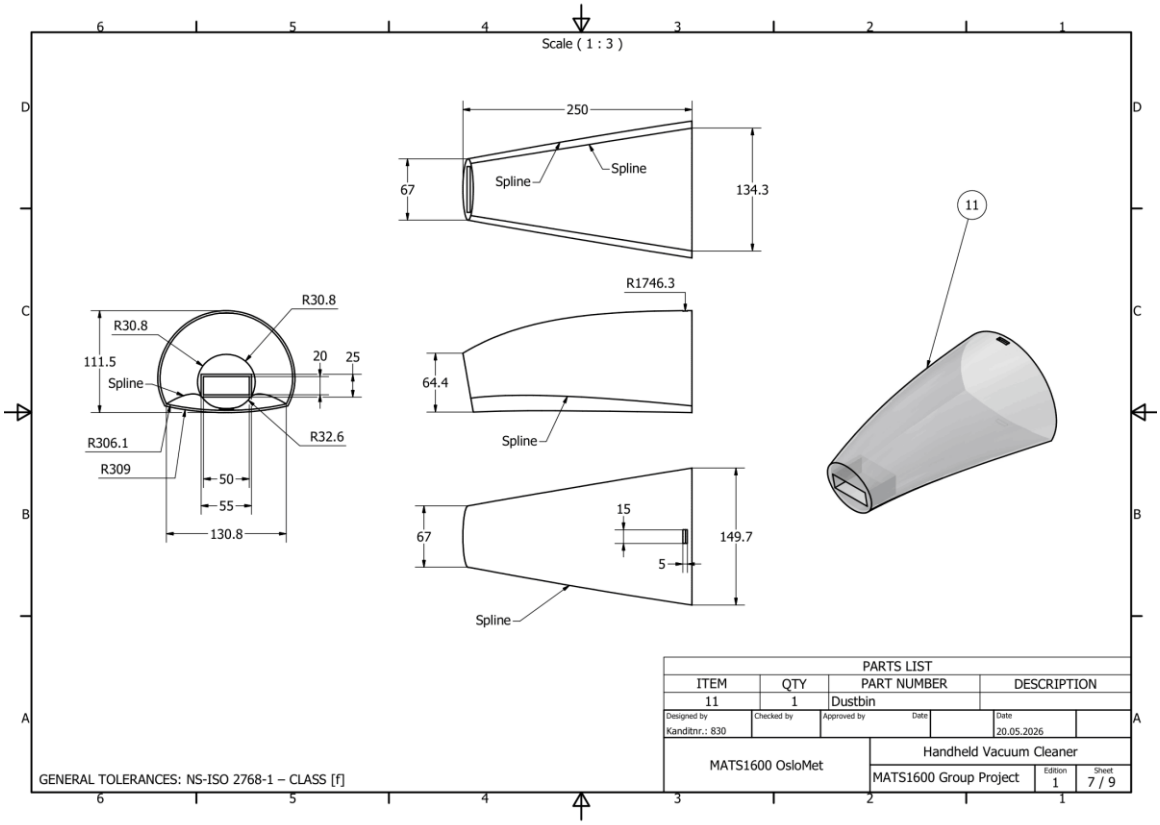
[14] 2D of motor



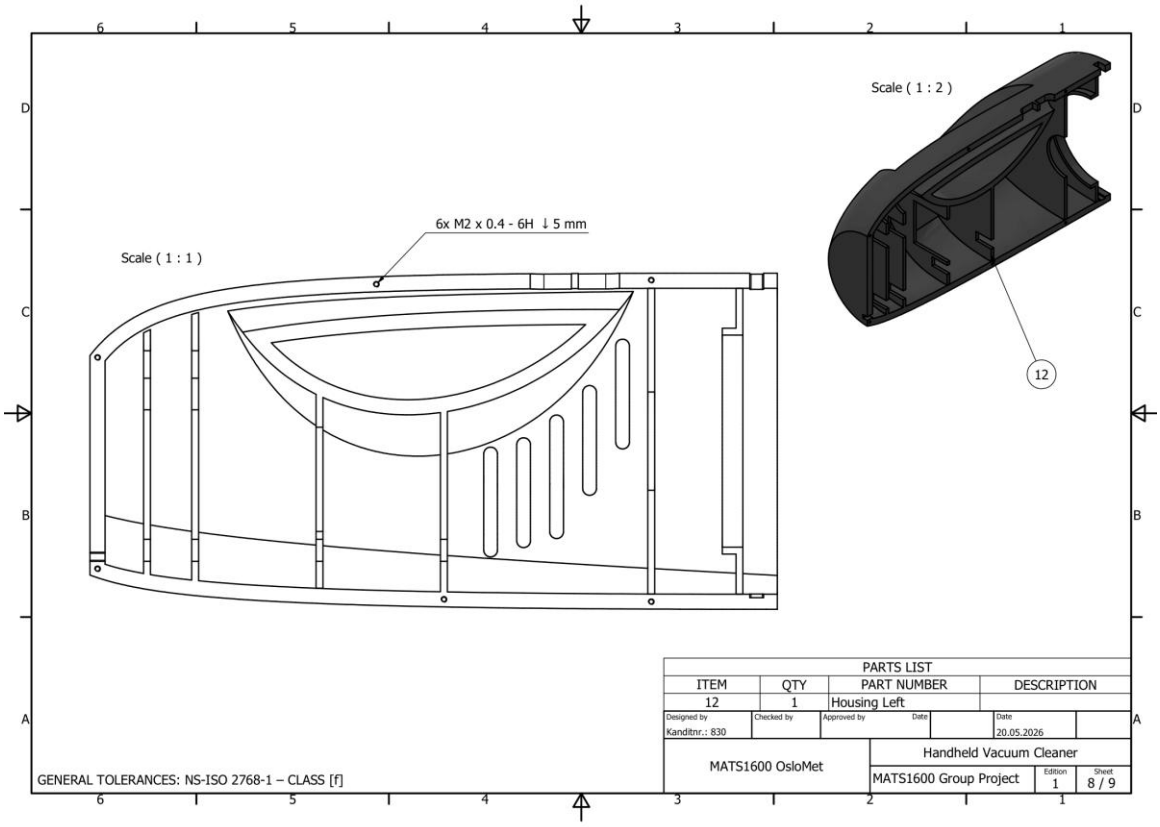
[15] 2D of tip


Scale (1 : 1)
98.5
50
42
12.3
4
2x R2
120°
42
50
4
12.3
20.3
4
9
PARTS LIST
ITEM
QTY
PART NUMBER
DESCRIPTION
9
1
Tip
Designed by
Kandiltnr.: 830
Checked by
Approved by
Date
Date
20.05.2026
MATS1600 OsloMet
Handheld Vacuum Cleaner
MATS1600 Group Project
Edition
1
Sheet
6 / 9
GENERAL TOLERANCES: NS-ISO 2768-1 – CLASS [f]

[16] 2D of Dustbin



[17] 2D of housing left



[18] 2D of housing right

